



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

B.Sc. CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	THEORETICAL CHEMISTRY II - GROUP THEORY AND MOLECULAR SPECTROSCOPY				
Type of Course	MAJOR				
Semester	V				
Academic Level	300 – 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	● Elementary awareness of degrees of freedom in molecules including translation, rotation and vibration. ● Basic understanding of quantum chemistry and the concept of quantised energy levels. ● Review the concept of electron spin introduced in the course, CHE3CJ201 ● Familiarity with the relationship between structure of molecules and their properties.				
Course Summary	The course introduces the theoretical aspects of two fundamental topics of theoretical chemistry which find a wide range of practical applications - Group theory and Molecular Spectroscopy. <i>Group theory</i> is a topic of mathematics which suggests that sets of elements (whatever be their nature) can be grouped into mathematical groups if they obey certain conditions. The course brings to light how group theory can be used for systematizing the study of structures of molecules based on their symmetry. This can be done by grouping the millions and millions of molecules that we come across into only a few mathematical groups called point groups based on the symmetry operations possessed by them. Such groupings help in simplifying the study of structure related properties of molecules.				

	Molecular Spectroscopy: The interaction of electromagnetic radiation with matter forms the basis for the different spectroscopic techniques used for the structural elucidation of molecules. The course brings to light the underlying principles involved in each of these spectroscopic techniques. The allowed energy states of molecules and the transitions between them are unique and are decided by the laws of quantum mechanics. Radiations having different frequencies (and hence different energy) interact with molecules and bring about characteristic transitions which help to identify the exact structure of molecules.
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Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO 1	<i>understand</i> the principles of constructing mathematical groups..	U	F	Assignment
CO 2	<i>realise</i> point groups as collections of symmetry operations of molecules	Ap	C	Class tests/Viva
CO 3	<i>identify</i> each spectroscopic method as the interaction of molecules with a characteristic radiation of the electromagnetic spectrum	An	P	Seminar/ Class tests
CO 4	<i>apply</i> various spectroscopic techniques for the structural elucidation of molecules.	Ap	P	Class tests/Assignment
CO 5	<i>justify</i> spectroscopic methods as unique tools for identifying molecules.	E	M	Viva
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks
I	Mathematical Preliminaries of Group Theory		6	12
	1	Conditions for sets of elements to form mathematical groups - closure rule, associativity, existence of identity and inverse elements.	2	

	2	Order of a group, Definitions of finite, infinite, Abelian and cyclic groups.	1	
	3	Binary combination of elements of a group - the group multiplication table and its features - general group multiplication tables of groups up to order 4.	2	
	4	Sub groups, similarity transformation and classes.	1	
	Sections from References: Section A			
II	Group Theory as a Means for the Systematic Study of Molecular Symmetry		12	25
	4	Tools for studying molecular symmetry – symmetry elements, symmetry operations and their classification.	2	
	5	Mathematical groups of symmetry operations - the point groups, nomenclature of point groups - Schoenflies notations.	2	
	6	Assigning point groups to molecules based on their symmetry elements.	2	
	7	Binary combinations of symmetry operations - the group multiplication tables of C_{2v} , C_{3v} and D_{2h} point groups.	3	
	8	Identifying classes of symmetry operations in point groups (C_{2v} and C_{3v} as examples)	2	
	9	Matrix representation of symmetry operations. Matrices for the symmetry operations of C_{2v} , C_{3v} and D_{2h} .	1	
	Sections from References: Section A			
III	Molecular Spectroscopy-I		18	36
	14	Energy levels in molecules – Born-Oppenheimer approximation. Electromagnetic spectrum - wavelength, frequency, wavenumber.	2	
	15	Interaction of electromagnetic radiation with matter - factors affecting line width and intensity of signal.	2	
	16	Rotational Spectroscopy: Introduction – Rigid rotor – Expression for energy – Selection rules – Intensities of spectral lines – Determination of bond lengths of diatomic molecules.	3	
	17	Vibrational Spectroscopy: Simple harmonic oscillator – Energy levels – Force constant – Selection rules - Anharmonicity – Fundamental frequencies – Overtones – Fingerprint region – Group frequency concept – Degree of freedom for polyatomic molecules – Modes of vibrations of CO_2 and H_2O .	4	

	18	Raman Spectroscopy: Basic principles – Rayleigh scattering - Raman scattering-Stokes & anti-stokes lines and their intensity difference - classical theory of Raman effect: polarizability - quantum theory of Raman scattering, selection rules for Raman spectra- Qualitative treatment of rotational Raman effect – Vibrational Raman spectra — Selection rules – Mutual exclusion principle. Resonance Raman scattering, Raman and IR spectroscopy,	4	
	19	Electronic Spectroscopy: Basic principles – Frank-Condon principle – Electronic transitions – Beer Lambert's law - Dissociation energy of diatomic molecules – Chromophore and auxochrome – Bathochromic and hypsochromic shifts.	3	
	Sections from References: Section B			
IV	Molecular Spectroscopy-II		12	25
	20	Nuclear Magnetic Resonance (NMR) Spectroscopy: Proton NMR spectroscopy – nuclei in a static magnetic field - basic principle of NMR spectroscopy - resonance; spectral parameters - chemical shift - nuclear shielding - spin-spin coupling - origin of coupling - coupling constants - NMR spectra of simple molecules.	6	
	21	¹³C NMR Spectroscopy: C-13 - relative abundance, chemical shift, spin-spin coupling. Factors affecting chemical shifts. Proton coupled and decoupled ¹³ C NMR.	3	
	22	Electron Spin Resonance (ESR) Spectroscopy: Principle - comparison between NMR and EPR - g factor - electron-nuclear interactions - hyperfine interactions – Hyperfine structure – ESR of methyl, phenyl and cycloheptatrienyl radicals.	3	
	Sections from References: Section B			
V	Open Ended Module: Learning through problem solving and plots		12	
	1	• Categorize molecules into point groups based on symmetry elements • Solving problems involving various spectroscopic data • Deducing the structure of various compounds from different spectra		

	Sections from References: Section A & Section B		
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Books and References:

Section A

1. F. A. Cotton, *Chemical Applications of Group Theory*, 3rd Edn., John Wiley & Sons, New York, 1990.
2. A. Salahuddin Kunju & G. Krishnan, *Group Theory & its Applications in Chemistry*, PHI Learning Pvt. Ltd. 2010.

Section B

1. C. N. Banwell, *Fundamentals of molecular spectroscopy*, McGraw-Hill, 1994.
2. P. W. Atkins, J. de Paula, *Atkin's Physical Chemistry*, 8th Edn., Oxford University Press 2006.
3. B. R. Puri, L. R. Sharma, M. S. Pathania, *Principles of Physical Chemistry*, 46. Edn., Vishal Publishing Company, New Delhi, 2013.
4. Donald A. McQuarrie, John D. Simon, *Physical Chemistry: A Molecular Approach*, University Science Books: Sausalito, CA; 1997.

Further reading

1. K. Veera Reddy, *Symmetry & Spectroscopy of Molecules* 2nd Edn., New Age International 2009.
2. H. H. Jaffe and M. Orchin, *Symmetry in Chemistry*, John Wiley & Sons Inc., 1965.
3. G. M. Barrow, *Introduction to Molecular Spectroscopy*, McGraw Hill, London, 1962.
4. Thomas Engel, *Quantum Chemistry & Spectroscopy*, Pearson education, 2006.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	1	-	-	-	-	-	3				1		2
CO 2	2	3	-	-	-	-	3						2
CO 3	-	-	1	-	-	-	3				2		1

CO 4	-	-	2	3	-	-	3			2	3		2
CO 5	-	1	-	-	-	-	3		1	2	2		2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment/viva/seminar	Practical skill Evaluation	End Semester Examinations
CO 1		✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5		✓		✓